

Emission-Aware Stochastic Resource Planning Scheme for Data Center Microgrid Considering Batch Workload Scheduling and Risk Management

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Abstract—Internet data centers are booming in the last few years, leading to gigantic amount of energy consumption and greenhouse gas emission. In this paper, a data center microgrid including conventional units, energy storage, and renewable generation is modeled to optimize its electricity bill and carbon footprint. Considering the randomness of electricity price, renewable output and workload distribution, the volatility risk is incorporated to measure the associated operation risk. Then, a day-ahead emission-aware stochastic resource planning scheme is formulated to decide the strategy on power procurement, energy storage operation, batch workload allocation, and unit commitment of conventional units. Simulation results show the effectiveness of the proposed approach.

Index Terms—Carbon emission, data center, risk management, stochastic optimization.

NOMENCLATURE

A. Indices and Sets

t, T Index and total number of time slots.
 i, I Index and total number of interactive workloads.
 j, J Index and total number of batch workloads.
 l, L Index and total number of conventional units.
 s, S Index and total number of stochastic scenarios.

B. System Parameters

$\lambda_{t,s}^i$ Amount of i th interactive workload at time t and scenario s .
 $w_{t,s}^i$ Total workload demand at time t and scenario s .
 B_{IT} Data center IT capacity limitation.
 D Maximum delay bound for all workloads.

P_{peak} Power consumption of a busy server.
 P_{idle} Power consumption of an idle server.
 M Amount of servers in a data center.
 PUE Power usage effectiveness of the data center.
 ε, θ Auxiliary variables in calculation.
 ES_t Energy storage level at the beginning of time t .
 η_c Charging efficiency of ESS.
 η_d Discharging efficiency of ESS.
 ES_{max} Maximum storage level of ESS.
 ES_{min} Minimum storage level of ESS.
 P_{max}^{char} Maximum charging rate of ESS.
 $P_{max}^{dischar}$ Maximum discharging rate of ESS.
 O_l No load cost of l th conventional unit.
 C_l Marginal cost of l th conventional unit.
 SU_l Start-up cost of l th conventional unit.
 SD_l Shut-down cost of l th conventional unit.
 $P_{l,max}^{unit}$ Maximum power output of l th conventional unit.
 $P_{l,min}^{unit}$ Minimum power output of l th conventional unit.
 RD_l Ramp-down rate of l th conventional unit.
 RU_l Ramp-up rate of l th conventional unit.
 MU_l Minimum-up time of l th conventional unit.
 MD_l Minimum-down time of l th conventional unit.
 e_k Carbon emission rate of fuel type k .
 $g_{k,t}$ Percentage of the marginal generation by fuel type k at time t .
 $P_{t,s}^{wind}$ Wind output at time t and scenario s .
 $P_{t,s}^{PV}$ Solar output at time t and scenario s .
 $\pi_{t,s}^{grid}$ Electricity price at time t and scenario s .
 $C_{t,s}^{grid}$ Cost of purchasing power from spot market at time t and scenario s .
 C_t^{unit} Operation cost of onsite conventional units at time t .
 E_t^{grid} Marginal carbon emission of grid power at time t .
 E_t^{unit} Carbon emission of internal units at time t .
 γ Carbon-emission coefficient.
 φ Risk coefficient.

C. Decision Variables

μ_t^j Allocated j th batch workload at time t .
 $Q_{t,s}$ Power consumption of data center at time t and scenario s .
 Z_t^{char} Binary variable to indicate if ESS is charging at time t .

Manuscript received February 25, 2018; revised June 4, 2018; accepted June 22, 2018. Date of publication June 28, 2018; date of current version October 12, 2018. Paper 2018-ESC-0109.R1, presented at the 2018 IEEE/IAS Industrial and Commercial Power Systems Technical Conference, Niagara Falls, ON, Canada, May 7–10, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Energy Systems Committee of the IEEE Industry Applications Society. This work supported in part by the National Key Research and Development Program of China (No. 2016YFE010240) and in part by Beijing Natural Science Foundation (3174057). (Corresponding author: Zhaohao Ding.)

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Digital Object Identifier 10.1109/TIA.2018.2851516

Z_t^{dischar}	Binary variable to indicate if ESS is discharging at time t .
P_t^{char}	Charging power of ESS at time t .
P_t^{dischar}	Discharging power of ESS at time t .
P_t^{grid}	Power purchased from grid at time t .
$P_{t,l}^{\text{unit}}$	Power output of l th conventional unit at time t .
$o_{t,l}$	Binary variable to indicate if l th conventional unit is on at time t .
$u_{t,l}$	Binary variable to indicate if l th conventional unit is starting up at time t .
$v_{t,l}$	Binary variable to indicate if l th conventional unit is shutting down at time t .
$\delta, \lambda(s)$	Auxiliary variables in conditional value at risk (CVaR).

I. INTRODUCTION

THE recent years have witnessed the rapid development of data centers all over the world. At the same time, the energy consumption and carbon emission of data centers are growing tremendously. According to [1], data centers in America consumed about 91 billion kWh electricity in 2013, which is about 2% of the total power consumption of the country in that year. Furthermore, it is predicted that data centers will attribute to around 8% of the global energy consumption and produce 340 metric megatons of carbon-dioxide emission per year by 2030.

Consequently, reducing the electricity bills and cutting the carbon footprint of data centers have drawn research attentions in the last few years. These studies can be roughly divided into several approaches [2]: adjust the switch state of servers, delay or drop a part of flexible workloads, use energy storage devices or renewable energy to cut the electricity fee, etc. Some of these studies also include co-optimization objective such as reducing the bandwidth cost or the operation cost of energy storage devices.

A number of literatures have been working on the energy management of data centers to minimize the operation cost or maximize profit. Luo *et al.* [3] proposed a novel energy cost optimization for data centers by dynamically scheduling workloads based on the time-varying electricity price. Ghamkhari and Mohsenian-Rad [4] approved that by combining the volatility of renewable energy and workloads, the electricity cost and carbon footprint would decrease significantly. Yu *et al.* [5] minimized the energy expenditure of data centers by scheduling battery and workload jointly while meeting the service delay guarantees. Chen *et al.* [6] considered the cooperation of IT system, power supply system, and cooling system to minimize the expected operational net-cost. However, few literatures have proposed a comprehensive approach including all these elements.

As environmental concerns have become one of the major priorities for most IT companies (i.e., data center operator or customer), the greenhouse gas emission optimization, such as carbon emission, is taken into consideration of data centers. Zhou *et al.* [7] put forward a three-way tradeoff problem of electricity cost, service requirement, and emission reduction budget for distributed data centers. Yu *et al.* [8] formulated the energy cost and carbon emission joint minimization problem in smart micro-

grid environments. Chen *et al.* [9] proposed day-ahead demand response programs for data centers to curtail cost and enhance environmental impacts. Most of the previous literatures adopted *average emission rate* to reflect the carbondioxide emitted by every kilowatt power generation. However, in the case of data center as a power consumer, *marginal emission rate* is a more accurate way to characterize the emission change due to the change of power demand [10]. In this paper, the marginal emissions of the utility grid and self-supply units are indicated by the weighted emission rates of marginal units in a reasonable manner.

The uncertainty factors such as dynamic workloads, volatile electricity price, and renewable generation may cause operation risk to a data center. In other words, the objective cost can be extremely high due to those uncertainties. To handle this challenge, Rao *et al.* [11] proposed a hedging scheme and formulated a bi-level program to minimize the operation risk. Yu *et al.* [12], [13] investigated the tradeoff between expected power cost/profit and operation risk depending on the risk preference level of data center operators. Ghamkhari *et al.* [14], [15] included operation risk constraints in their profit maximization problem. In this paper, conditional value at risk (CVaR) model [16] is adopted as a risk measurement strategy to adverse the upside deviation of expected operation cost.

Combining all the aforementioned components, it can be noted that limited attention had been drawn on co-optimization of available resources such as onsite generation resources, energy storage, utility grid procurement, and workload management while considering the stochastic nature of external factors. Meanwhile, the carbon footprint and risk management should also be included in the operation objective for data centers. Therefore, an emission aware stochastic resource planning scheme for data center microgrid is proposed in this paper.

The main contributions of this work compared with existed research are listed below. In this paper, an emission-aware and risk-constrained resource planning model for data center is proposed. The optimal unit commitment of onsite generators has been jointly optimized with the schedule of energy storage system (ESS) and batch workloads. The uncertainties of electricity market price, renewable generation, and workload distribution are fully considered and addressed by the proposed stochastic optimization model. Simulations based on historical data illustrate different resource planning schemes for data centers with different environment and risk preferences, which demonstrate the effectiveness of the proposed approach.

The rest of this paper is organized as follows. The mathematical models and optimal scheduling problem are formulated in Section III. Section IV provides numerical case study results and Section V presents the conclusion.

II. SYSTEM MODEL AND MATHEMATICAL FORMULATION

Nowadays, data centers are typically run in microgrid environments, which include conventional units, ESS, and distributed renewable generations [17], [18]. Consequently, the operation cost and carbon footprint are determined by the operator of data center microgrid. As shown in Fig. 1, energy procurement decision from utility grid and scheduling decision

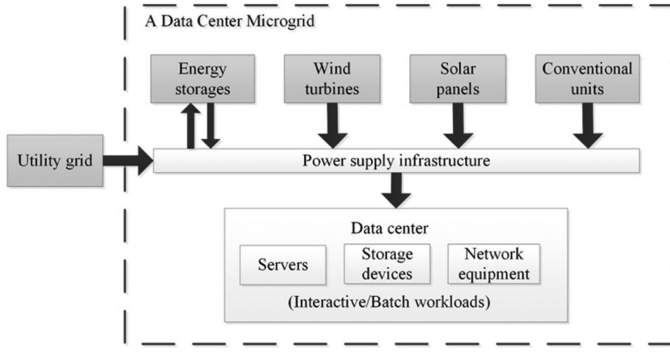


Fig. 1. Structure of a data center microgrid.

for onsite generation resources are co-optimized with the batch workload allocation decision to establish a reliable, economical, and environment-friendly resource planning scheme for the data center microgrid [19].

This paper takes a full consideration of the stochastic characteristics of the data center resource planning. The randomness of wholesale electricity market price, onsite renewable generation outputs, and workload distribution are captured by a set of scenarios, which are generated based on probabilistic distribution models.

A. Workload Allocation Model

Workloads arriving at data centers can be divided into two categories: interactive and batch workload [6]. The interactive workload, also known as delay-sensitive workload, must be served immediately (typically in a few seconds). And the batch workload, also called delay-tolerant workload, requires a flexible service time that can be delayed by hours. Generally speaking, batch jobs are normally consisted of data-intensive tasks such as scientific computing, image processing, and so on [20].

Many literatures have carried out detailed studies on the different characteristics of these two kinds of workloads. Some focus on the delay constraints of interactive jobs [15], [21], others pay attention to the flexible dispatch of batch jobs [5], [9]. In this paper, both types of workloads are considered. Specifically, batch workloads are optimally allocated to different time slots while the energy supply for all the workloads are scheduled based on the operation cost and carbon footprint.

Suppose that there are i types of interactive jobs and j types of batch jobs in total, each type may match a specific service requirement. Let $\lambda_{t,s}^i$ and μ_t^j represent the number of these two kinds of workloads at each time slot respectively, in which the first one is a stochastic parameter. For simplicity, the number of workload is expressed by the computing resource needed. And the CPU resource is seemed as the only limiting factor of workload processing. So the overall amount of interactive and batch jobs at time t and scenario s must not exceed the maximum CPU capacity, which is defined in (2) as follows:

$$w_{t,s} = \sum_i \lambda_{t,s}^i + \sum_j \mu_t^j \quad \forall t, s \quad (1)$$

$$0 \leq w_{t,s} \leq B_{IT} \quad \forall t, s. \quad (2)$$

To guarantee the quality of service of data centers, there will be a limitation of the longest defer time. Once the threshold is surpassed, all the unfinished batch jobs will be dropped. The maximum allowed completion time is denoted as follows:

$$\max(\Delta t_j) \leq D \quad \forall j. \quad (3)$$

Except for supporting IT devices like servers and hard-drive storage, data centers also consume a big part of energy for the cooling system, room lighting, and electricity distribution network for other loads. Those are necessary to maintain the normal operation of a data center. Power usage effectiveness (PUE), which is defined as the ratio of total power consumption of a data center to the power consumed by IT-related equipment [22], is adopted by this paper to measure the energy usage efficiency. In this way, the power consumption of a data center can be expressed as shown in [21]

$$Q_{t,s} = M[\varepsilon + \theta w_{t,s}] \quad \forall t, s \quad (4)$$

$$\varepsilon \triangleq P_{idle} + (PUE - 1)P_{peak} \quad (5)$$

$$\theta \triangleq P_{peak} - P_{idle}. \quad (6)$$

It should be noted that the above formulas are derived under homogeneous conditions, but it is also suitable for heterogeneous cases with a few additional notations.

B. Power Supply Model

The power requirement of a data center can be met by the bulk power system or the onsite generators and energy storage devices, depending on the operational conditions.

The ESS is a prevalent component of practical microgrids for the purpose of ensuring power supply reliability and cutting down the volatility of renewable energy. As there is conversion loss in the process of power charging and discharging, charging and discharging efficiency are indicated in (7). The charge and discharge power at each time slot should be both non-negative and limited within the upper bound. Two binary variables are introduced to indicate the charge and discharge process. And it is reasonable to assume that the two processes should not happen at the same time. Therefore, the following equations and constraints can be formed to model the ESS:

$$ES_{t+1} = ES_t + \eta_c P_t^{\text{char}} - \eta_d P_t^{\text{dischar}} \quad \forall t \quad (7)$$

$$ES_{\max} \leq ES_t \leq ES_{\min} \quad \forall t \quad (8)$$

$$Z_t^{\text{char}}, Z_t^{\text{dischar}} \in \{0, 1\} \quad \forall t \quad (9)$$

$$Z_t^{\text{char}} + Z_t^{\text{dischar}} \leq 1 \quad \forall t \quad (10)$$

$$0 \leq P_t^{\text{char}} \leq Z_t^{\text{char}} P_{\max}^{\text{char}} \quad \forall t \quad (11)$$

$$0 \leq P_t^{\text{dischar}} \leq Z_t^{\text{dischar}} P_{\max}^{\text{dischar}} \quad \forall t. \quad (12)$$

In previous works [6], [17], the model of conventional units in data center microgrid has been simplified to the maximum output limitation and ramping constraints. Furthermore, only the economic dispatch operation cost is modeled while costs associated with unit commitment process, such as start-up and shut-down cost of units, are neglected.

In this paper, a unit commitment operation cost model for conventional generators is established, as shown in the following:

$$C_t^{\text{unit}} = \sum_{t=1}^T \sum_{l=1}^L [\text{SU}_l u_{t,l} + \text{SD}_l v_{t,l} + O_l o_{t,l} + C_l P_{t,l}^{\text{unit}}]. \quad (13)$$

The constraints of unit commitment decision-making process are listed as follows [23], [24]. The capacity constraint of conventional generators is indicated in (15). The ramp-up and ramp-down time requirements for each generator are described in (16) and (17). Ramping rate constraints are formulated in (18)–(21):

$$P_t^{\text{unit}} = \sum_{l=1}^L P_{t,l}^{\text{unit}} \quad \forall t \quad (14)$$

$$P_{l \min}^{\text{unit}} o_{t,l} \leq P_{t,l}^{\text{unit}} \leq P_{l \max}^{\text{unit}} o_{t,l} \quad \forall t, l \quad (15)$$

$$-o_{t-1,l} + o_{t,l} - o_{k,l} \leq 0 \quad 2 \leq k - (t-1) \leq \text{MU}_l \quad \forall t, l \quad (16)$$

$$-o_{t-1,l} - o_{t,l} + o_{k,l} \leq 1 \quad 2 \leq k - (t-1) \leq \text{MD}_l \quad \forall t, l \quad (17)$$

$$-o_{t-1,l} + o_{t,l} - u_{t,l} \leq 0 \quad \forall t, l \quad (18)$$

$$o_{t-1,l} - o_{t,l} - v_{t,l} \leq 0 \quad \forall t, l \quad (19)$$

$$P_{t,l}^{\text{unit}} - P_{t-1,l}^{\text{unit}} \leq (2 - o_{t-1,l} - o_{t,l}) P_{l \min}^{\text{unit}} + (1 + o_{t-1,l} - o_{t,l}) \text{RU}_l \quad \forall t, l \quad (20)$$

$$P_{t,l}^{\text{unit}} - P_{t-1,l}^{\text{unit}} \leq (2 - o_{t-1,l} - o_{t,l}) P_{l \min}^{\text{unit}} + (1 - o_{t-1,l} + o_{t,l}) \text{RD}_l \quad \forall t, l. \quad (21)$$

Combining the supply side and demand side together, the power balance constraint can be obtained as follows:

$$P_t^{\text{grid}} + P_t^{\text{unit}} + P_{t,s}^{\text{wind}} + P_{t,s}^{\text{PV}} + Z_t^{\text{dischar}} P_t^{\text{dischar}} - Z_t^{\text{char}} P_t^{\text{char}} = Q_{t,s} \quad \forall t, s. \quad (22)$$

As the electricity price in spot market varies in different time slots and scenarios, the power procurement cost can be expressed by

$$C_{t,s}^{\text{grid}} = P_t^{\text{grid}} \pi_{t,s}^{\text{grid}} \quad \forall t, s. \quad (23)$$

C. Carbon Emission Model

The power consumed by a data center, whether provided by the grid or the onsite generators, will emit carbon dioxide simultaneously. The carbon emission rates of different fuel types are calculated based on the data and method provided by EIA [25]. For instance, the emission factors of coal-fired and gas-fired units are 979 and 418 kg/MWh, respectively, and renewables are carbon free while generating power.

Assuming that the data center resource scheduling decisions will not change the operation decisions of the utility grid, the carbon footprint impact of data center resource planning scheme on the bulk power system is modeled by marginal carbon emission

rate of each operation interval. The marginal carbon emission of bulk power system can be obtained as

$$E_t^{\text{grid}} = P_t^{\text{grid}} \sum_{k=1}^n e_k g_{k,t}. \quad (24)$$

In this paper, the percentage of marginal generation by fuel types are derived from the PJM real-time market data, which is published regularly on the PJM website [26].

The formula above is also applicable for the emission of internal units, which is denoted by

$$E_t^{\text{unit}} = \sum_{l=1}^L e_l P_{t,l}^{\text{unit}}. \quad (25)$$

D. Operation Risk Model

In existing works, two main risk metrics are introduced: mean variance [11], [14], [27] and CVaR [12], [13], [21]. The mean variance method can cut down the fluctuation of operation costs, while a data center pays more attention on avoiding excessively high costs. Therefore, CVaR method is adopted to measure the operation risk in this paper.

The CVaR of a random variable at the confidence level α is defined as the expected value of the $(1 - \alpha) \times 100\%$ scenarios in the worst case [12]. It can be formulated as

$$\text{CVaR}_\alpha(A(s)) = \min \left\{ \delta + \frac{1}{1 - \alpha} \sum_{s=1}^{NS} \omega(s) \lambda(s) \right\} \quad (26)$$

s.t.

$$A(s) - \delta - \lambda(s) \leq 0 \quad (27)$$

$$\lambda(s) \geq 0 \quad \forall s. \quad (28)$$

E. Minimization Problem

Combining all those aforementioned factors together, data center operators need to determine the tradeoffs among the energy cost by purchasing power or self-generating, the carbon emission caused by the grid power indirectly or onsite generators directly, and the operation risk due to stochastic variables. In the proposed production cost model, the marginal cost of renewable energy and battery degradation is neglected, as well as their carbon emission. The three parts are weighted differently according to diverse environmental and risk preference of data center operators.

So the optimization objective of the proposed emission-aware stochastic resource schedule can be expressed as an expected value

$$\min \sum_{t=1}^T \mathbb{E} \{ [C_{t,s}^{\text{grid}} + C_t^{\text{unit}}] + \gamma [E_t^{\text{grid}} + E_t^{\text{unit}}] + \varphi \text{CVaR} \} \quad (29)$$

where

$$C_{t,s}^{\text{grid}} = P_t^{\text{grid}} \pi_{t,s}^{\text{grid}} \quad (30)$$

$$C_t^{\text{unit}} = \sum_{t=1}^T \sum_{l=1}^L [\text{SU}_l u_{t,l} + \text{SD}_l v_{t,l} + O_l o_{t,l} + C_l P_{t,l}^{\text{unit}}] \quad (31)$$

$$E_t^{\text{grid}} = P_t^{\text{grid}} \sum_{k=1}^n e_k g_{k,t} \quad (32)$$

$$E_t^{\text{unit}} = \sum_{l=1}^L e_l P_{t,l}^{\text{unit}} \quad (33)$$

$$\text{CVaR}_\alpha(A(s)) = \min \left\{ \delta + \frac{1}{1-\alpha} \sum_{s=1}^{\text{NS}} \omega(s) \lambda(s) \right\} \quad (34)$$

s.t.

$$w_{t,s} = \sum_i \lambda_{t,s}^i + \sum_j \mu_t^j \quad \forall t, s \quad (35)$$

$$0 \leq w_{t,s} \leq B_{\text{IT}} \quad \forall t, s \quad (36)$$

$$\text{ES}_{t+1} = \text{ES}_t + \eta_c P_t^{\text{char}} - \eta_d P_t^{\text{dischar}} \quad \forall t \quad (37)$$

$$\text{ES}_{\text{max}} \leq \text{ES}_t \leq \text{ES}_{\text{min}} \quad \forall t \quad (38)$$

$$Z_t^{\text{char}}, Z_t^{\text{dischar}} \in \{0, 1\} \quad \forall t \quad (39)$$

$$Z_t^{\text{char}} + Z_t^{\text{dischar}} \leq 1 \quad \forall t \quad (40)$$

$$0 \leq P_t^{\text{char}} \leq Z_t^{\text{char}} P_{\text{max}}^{\text{char}} \quad \forall t \quad (41)$$

$$0 \leq P_t^{\text{dischar}} \leq Z_t^{\text{dischar}} P_{\text{max}}^{\text{dischar}} \quad \forall t \quad (42)$$

$$P_t^{\text{unit}} = \sum_{l=1}^L P_{t,l}^{\text{unit}} \quad \forall t \quad (43)$$

$$P_{l,\text{min}}^{\text{unit}} o_{t,l} \leq P_{t,l}^{\text{unit}} \leq P_{l,\text{max}}^{\text{unit}} o_{t,l} \quad \forall t, l \quad (44)$$

$$-o_{t-1,l} + o_{t,l} - o_{k,l} \leq 0 \quad \forall t, l \quad (45)$$

$$2 \leq k - (t-1) \leq \text{MU}_l \quad \forall t, l \quad (46)$$

$$-o_{t-1,l} - o_{t,l} + o_{k,l} \leq 1 \quad \forall t, l \quad (47)$$

$$2 \leq k - (t-1) \leq \text{MD}_l \quad \forall t, l \quad (48)$$

$$-o_{t-1,l} + o_{t,l} - u_{t,l} \leq 0 \quad \forall t, l \quad (49)$$

$$o_{t-1,l} - o_{t,l} - v_{t,l} \leq 0 \quad \forall t, l \quad (50)$$

$$P_{t,l}^{\text{unit}} - P_{t-1,l}^{\text{unit}} \leq (2 - o_{t-1,l} - o_{t,l}) P_{l,\text{min}}^{\text{unit}} + (1 + o_{t-1,l} - o_{t,l}) \text{RU}_l \quad \forall t, l \quad (51)$$

$$P_{t,l}^{\text{unit}} - P_{t-1,l}^{\text{unit}} \leq (2 - o_{t-1,l} - o_{t,l}) P_{l,\text{min}}^{\text{unit}} + (1 - o_{t-1,l} + o_{t,l}) \text{RD}_l \quad \forall t, l \quad (52)$$

$$P_t^{\text{grid}} + P_t^{\text{unit}} + P_{t,s}^{\text{wind}} + P_{t,s}^{\text{PV}} + Z_t^{\text{dischar}} P_t^{\text{dischar}} - Z_t^{\text{char}} P_t^{\text{char}} = Q_{t,s} \quad \forall t, s \quad (53)$$

As shown above, γ and φ are the carbon-emission and risk coefficients, respectively. When the coefficients are set to 0, the carbon emission and operation risk are not taken into account.

TABLE I
PARAMETERS OF DATA CENTER

Maximum power consumption(MW)	Number of servers	Pidle (W)	Ppeak (W)	PUE
50	2.5×10^5	100	200	1.2

TABLE II
PARAMETERS OF ESS

Maximum state(MWh)	Minimum state(MWh)	Initial and final state(MWh)	Maximum charging/discharging rate (MW/h)	Charging/discharging efficiency
30	5	5	5	0.8

With their value increasing, the data center operator will attach more importance to environmental impact and risk averseness.

It should be noted that the decision variables for this stochastic optimal problem include the allocation of batch workloads, the operation schedule of ESS and conventional units, as well as the power purchased from electricity market. The proposed resource planning model is formulated as a mixed-integer linear programming (MILP) problem.

The sample average approximation method is applied to solve the stochastic features of electricity price, renewable energy, and interactive workloads. The true distribution of these random variables is approximated by Monte Carlo sampling technology with an empirical distribution. The probability of each scenario s is equal to $1/S$. Therefore, the random parameters in the objective function can be replaced by the following:

$$\min \frac{1}{S} \sum_{s=1}^S \sum_{t=1}^T \left\{ \left[C_{t,s}^{\text{grid}} + C_t^{\text{unit}} \right] + \gamma \left[E_t^{\text{grid}} + E_t^{\text{unit}} \right] + \varphi \text{CVaR} \right\}. \quad (52)$$

III. CASE STUDIES

A sample data center microgrid system is established to examine the proposed model in this section. The formulation is coded in MATLAB [28] and CPLEX solver [29] is employed to solve the MILP problem. All the models are implemented on a computer with AMD X6 CPU@2.70 GHz and 8 GB memory.

A. Simulation Setup

In this paper, the proposed data center microgrid includes three conventional generators (two gas based and one coal based), distributed wind and solar generations, and ESS. The characteristics of data center, ESS, and onsite conventional units are listed in Table I.

The parameters for data center are obtained based on [21] and [17]. The PUE of data center is set to be 1.2 according to [30]. The characteristics of conventional generation units and ESS are scaled from ERCOT scheduling data [23], [24].

The carbon emission rate of the bulk power system is calculated based on the historical hourly marginal unit mix data of

TABLE III
PARAMETERS OF CONVENTIONAL UNITS

Unit	Fuel Type	High/Low Sustainable Limit (MW)	Ramp Up/Down Rate (MW/h)	Min Up/Down Time (h)
Unit1	Gas	15/5	4	4
Unit2	Gas	20/5	3.5	4
Unit3	Coal	20/9	6	3

TABLE IV
PARAMETERS OF CONVENTIONAL UNITS

Unit	Initial state	Initial power (MW)	Start-up/shut down cost (\$)	No load cost (\$)	Marginal cost (\$/MWh)
Unit1	On	10	50	40	18
Unit2	On	12	55	37	18
Unit3	On	14	40	30	16

TABLE V
CARBON EMISSION RATE OF DIFFERENT FUEL TYPES

Fuel Type	Coal	Gas	Wind	Solar	Nuclear
Emission rate (kg/MWh)	979	418	0	0	0

TABLE VI
PARAMETERS OF BATCH WORKLOADS

Workload	Arrival time	Deadline
Batch job 1	0:00	24:00
Batch job 2	0:00	12:00
Batch job 3	13:00	24:00

TABLE VII
UNCERTAINTIES OF THE RENEWABLE GENERATION

Type	Nominal Rating(MW)	Normalized Standard Deviation of Day-ahead Forecasting Error
Wind	5	13.58%
Solar	3	5.10%

PJM [26]. And the emission rates of each fuel type are listed in Table V [25], [31].

The interactive workload distribution is normalized based on the historical workload information in [3]. In this paper, three kinds of batch jobs are assumed based on [18]. Without loss of generality, batch jobs 1 and 2 are assumed to arrive at midnight, while job 3 arrives at noon, as shown in Table VI. In all, 50% of the workloads are assumed to be interactive workloads and the other 50% are batch jobs.

The wind and solar generation data are obtained from a pilot project in North China. The normalized standard deviation of the day-ahead forecasting error is obtained from research [32], [33], as shown in Table VII.

TABLE VIII
PARAMETER SETTINGS OF THREE CASES

Case	γ	φ
Case 1	0.0025	0
Case 2	80	0
Case 3	0.01	0

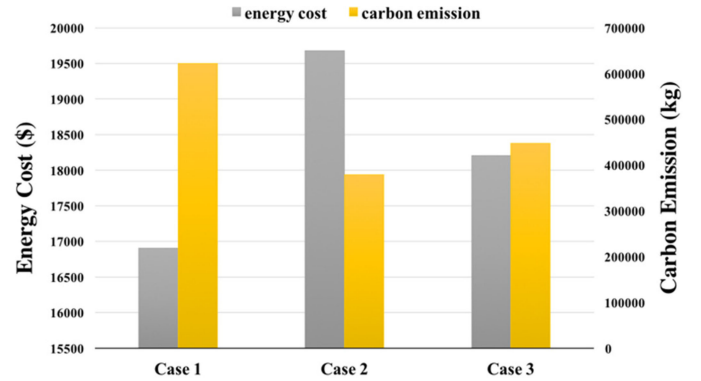


Fig. 2. Cost and emission comparison for three cases.

B. Numerical Results Analysis

In this section, the sample data center is explored in details. As shown in Table VIII, three cases are proposed to illustrate the tradeoff between energy cost and carbon emission in the proposed stochastic resource planning scheme. The risk coefficient is set to zero (i.e., risk neutral) for all three cases to eliminate the effect of risk preference.

Case 1 is set as cost-driven planning case. In this case, the major purpose of resource scheduling is to minimize energy cost, including power procurement and self-supply cost. This case is aimed to simulate the behavior of data centers aimed to gain the most economic benefit.

Case 2 is set as emission-driven planning case. In this case, the main objective of resource scheduling is to minimize carbon emission from both utility grid and internal generation. This case is designed to simulate the behavior of data centers aimed to operate in the most environment-friendly way.

Case 3 is set as cost-emission multiobjective planning case. In this case, both energy cost and carbon emission are taken into consideration in the resource scheduling objective. This case is designed to simulate the behavior of data centers aimed to operate in a multiobjective manner to minimize the energy cost and carbon footprint simultaneously.

As shown in Fig. 2, the energy cost is the lowest in Case 1, while the carbon emission is the lowest in Case 2. Case 3 presents a relatively balanced result with medium values for both the energy cost and carbon emission. The results demonstrate that the selection of emission coefficient could significantly affect the portfolio of energy cost and carbon emission for the resource planning of data center. In practice, the data center operators can determine the emission coefficient based on its preference on carbon emission. For example, an environmental driven data center, such as SAP data center in St.

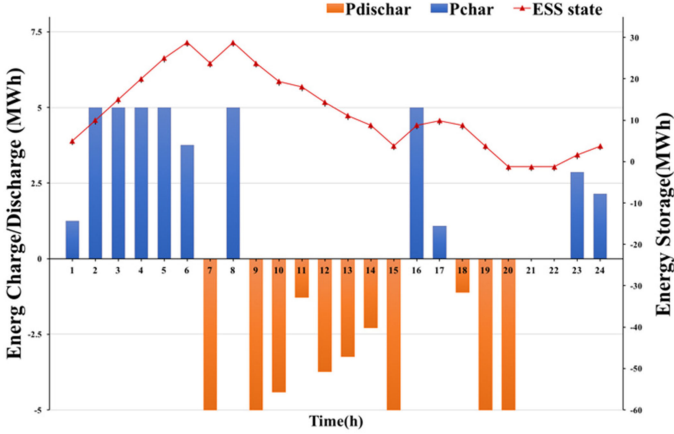


Fig. 3. ESS schedule decisions in case 1.

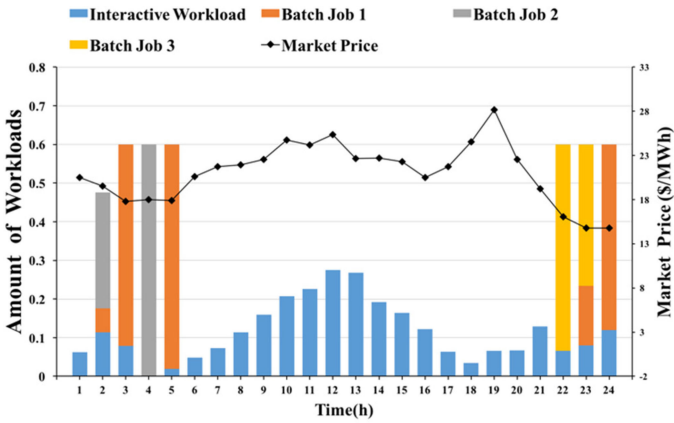


Fig. 4. Workload allocation decisions in case 1.

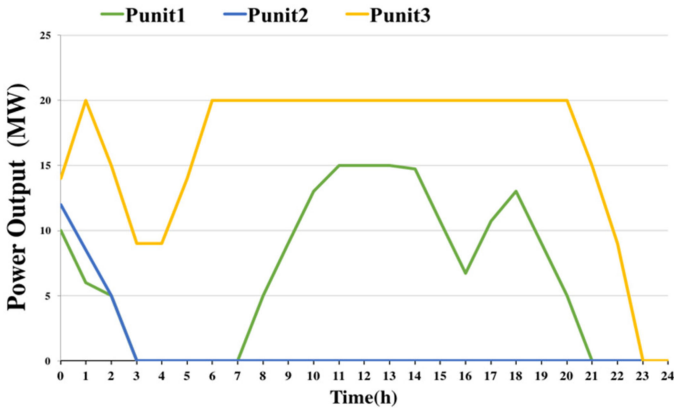


Fig. 5. Unit commitment decisions in case 1.

Leon-Rot Germany [34], may choose relatively large value for its carbon-emission coefficient in the resource planning model.

The day-ahead schedule for case 1 is presented in Figs. 3, including the schedule of ESS, batch jobs, conventional unit commitment, and the overall resource planning scheme. Aiming at reducing energy cost, the wholesale electricity price is also presented in Figs. 4 and 6 as a reference.

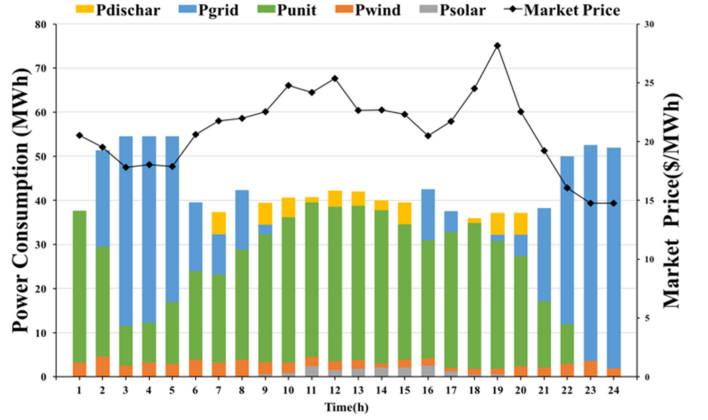


Fig. 6. Day-ahead resource scheduling decisions in case 1.

As shown in Fig. 3, in the cost-driven case, the key factor of energy storage schedule is the electricity price. Energy storage is encouraged to charge when the electricity price is low and discharge at high prices. As the result illustrates, the energy storage is arranged to charge when the price is relatively low (1:00–6:00, 16:00–17:00, 23:00–24:00), discharge at higher points (9:00–15:00, 18:00–20:00).

Fig. 4 depicts the allocation of workloads. The underneath blue bars represent the amount of interactive workloads in every time slot, which must be solved in time. Batch job 2 is allocated to 2:00–4:00, among the lowest price interval between 0:00–12:00. Likewise, batch job 3 is allocated between 22:00–23:00, the lowest price interval between 13:00–24:00. As batch job 1 can be allocated during the whole day, it is distributed into two time periods (2:00–5:00, 23:00–24:00).

As demonstrated by Fig. 5, for the unit commitment decision, the contribution of unit 3 is significantly larger than units 1 and 2. This can be explained by the fact that unit 3 has the lowest marginal operational cost. When the electricity price is relatively low (2:00–5:00, 22:00–24:00), most of the power is provided by the power market. So the conventional units should decrease their output, even shut down if necessary. While the start-up cost of unit 1 is smaller than unit 2, it has priority to start again. It can be noticed that unit 2 shuts down at 3:00 and does not start up during the whole day, then units 1 and 3 also shut down when the day ends.

The optimal resource planning scheme for case 1 is depicted in Fig. 6. The greatest power contribution during the day comes from conventional units, because they are initially running and are mainly dispatched when the electricity price is higher than marginal fuel costs. When the electricity price is relatively high (6:00–20:00), most power is provided by onsite generators and ESS discharging energy. When the price is lower (2:00–5:00, 21:00–24:00), the power consumption of the data center is higher (due to more batch jobs are distributed to these time slots), and a higher proportion is purchased from the electricity market. At the bottom of this figure, renewable generation comprises a small proportion of power consumption all over the day, limited to their installed capacity.

The day-ahead schedule for case 2 is presented in Figs. 7. As case 2 aims at minimizing the carbon emission of power

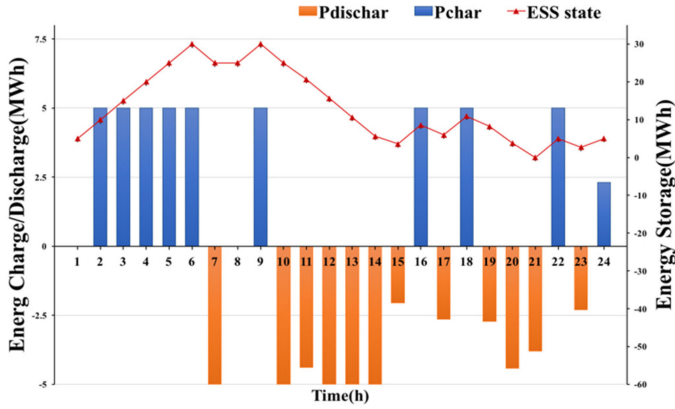


Fig. 7. ESS schedule decisions in case 2.

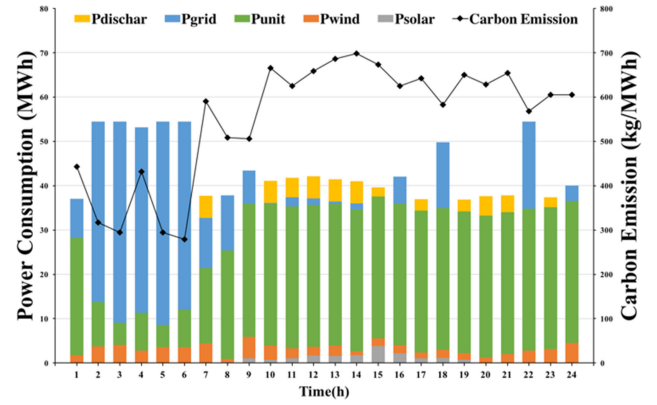


Fig. 10. Day-ahead resource scheduling decisions in case 2.

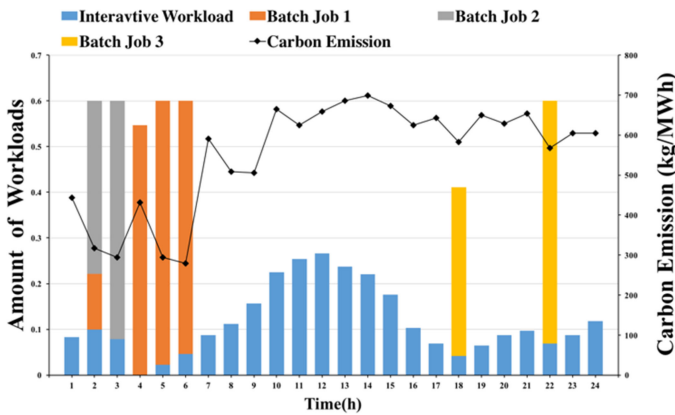


Fig. 8. Workload allocation decisions in case 2.

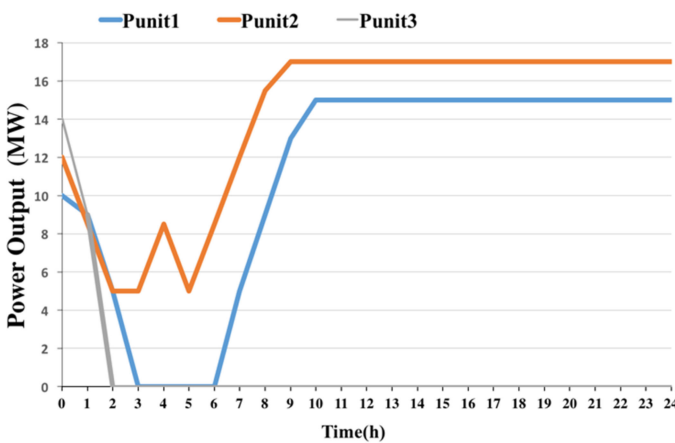


Fig. 9. Unit commitment decisions in case 2.

consumption, the carbon emission rate of utility grid (instead of the spot market price) is presented in Figs. 8 and 10.

As demonstrated in Fig. 7, in the emission-driven case, the key factor of energy storage schedule is carbon emission rate of the grid. Energy storage is encouraged to charge when the carbon emission of grid is lower than conventional units and discharge at high emission rates. As the result illustrates, the energy storage is arranged to charge when the emission rate

of grid is relatively low (2:00–6:00, 9:00, 16:00, 18:00, 22:00), discharge at higher points (7:00, 10:00–15:00, 19:00–21:00).

Fig. 8 depicts the allocation of workloads in case 2. Similarly, the underneath blue bars represent the amount of arrived interactive workloads in every time slot, which must be solved in time. Batch job 2 is allocated to 2:00–3:00, among the lowest emission interval between 0:00–12:00. Batch job 3 is allocated to 18:00 and 22:00, two lowest points of emission curve between 13:00–24:00. As batch job 1 can be allocated during the whole day, it is distributed to the lowest emission intervals (2:00–6:00). Both the ESS and batch job allocation can be verified in Fig. 10.

As for conventional unit commitment in Fig. 9, the decision making depends on the comparison between gas/coal units and the marginal emission rate of grid. Since the carbon emission rate of coal-fired generators (979 kg/MWh) is much higher than gas-fired generators (418 kg/MWh), unit 3 shuts down directly at the very start. When the emission rate of grid is lower than onsite gas-fired generators (1:00–6:00), most of the power is provided by the power market. So the two gas-fired units should decrease their output. Though the emission rate of units 1 and 2 are the same, the shut-down cost of unit 1 is smaller than unit 2. Therefore, when it is necessary unit 1 should be shut down first. It can be noticed that unit 1 shuts down at 3:00 and starts up again at 6:00, when the carbon emission of grid increased to a higher level.

The optimal resource planning scheme for case 2 is depicted in Fig. 10. The greatest power contribution during the day comes from gas units, because their emission rate is much lower than the marginal emission rate of grid at most time, which can be explained by the large percentage of coal-fired units in the grid resource mix. It can be seen that when the grid emission rate is higher than gas units (7:00–24:00), most power is provided by gas units and ESS discharging energy. When grid emission rate is relatively lower (2:00–6:00), the power consumption of data center is higher (due to batch jobs are distributed to these time slots), and most power is obtained from the electricity market. At the bottom of this figure, the renewable generation comprises a small proportion of power consumption all over the day, limited to their installed capacity.

To evaluate the tradeoff between the cost and operation risk, simulations are conducted based on case 1 and the risk

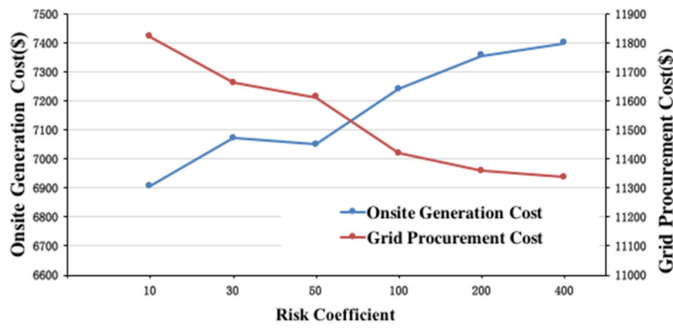


Fig. 11. Impact of changing risk coefficient.

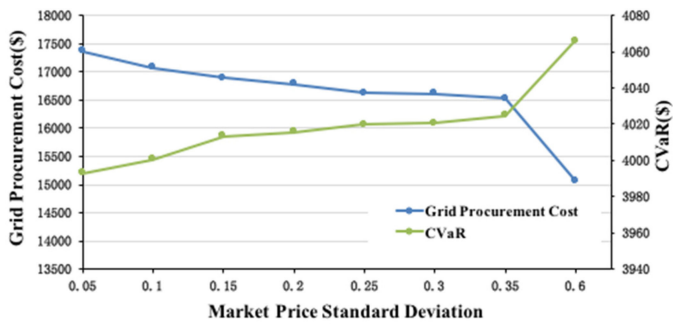


Fig. 12. Impact of forecast error on grid procurement cost and operation risk.

coefficient is investigated. As shown in Fig. 11, the risk coefficient is changed to illustrate the effect of risk preference on grid procurement cost and onsite generation cost. As the risk coefficient increasing, the data center becomes more and more risk-averse. As the volatility associated with market introducing more uncertainties, the grid procurement cost decreases and the generation cost of onsite units increases with the risk coefficient growing.

To illustrate the impact of stochastic variables on CVaR and schedule results, the electricity market price is chosen to be analyzed in detail. The impact of spot price forecasting error (i.e., spot price standard deviation) on grid procurement cost and operation risk is shown in Fig. 12. It can be observed that with the decrease of forecasting accuracy, the value of CVaR (the operation risk) increases and less power are bought from utility grid, which leads to the reduction of grid procurement cost.

IV. CONCLUSION

A stochastic resource planning scheme for data center is proposed to jointly optimize the energy cost, carbon emission, and operation risk in this paper. The stochastic characteristics of electricity market price, renewable generation, and workload distribution are taken into account. The proposed problem is formulated as a MILP problem. The schedule of power procurement, energy storage operation, conventional unit commitment, and batch workload allocation are obtained in different premium of environmental concerns. Also, the effect of risk preference and forecast accuracy is discussed. The results illustrated

the tradeoff relationships among three parts in the objective function and the effectiveness of proposed scheme is shown.

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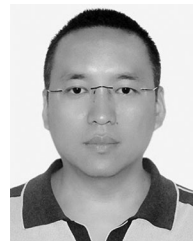
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